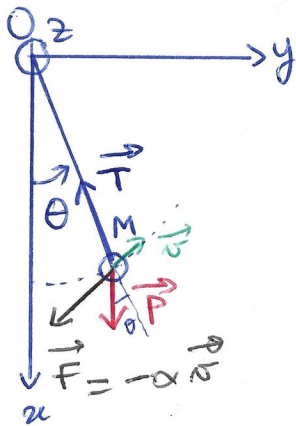


Pendule avec frottements fluide

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2 ① Bilan:

- poids $\vec{P}$
- tension $\vec{T}$
- force de frotts $\vec{F}$.

$$\vec{m}_0(\vec{T}) = \vec{OM} \wedge \vec{T} = \underline{\underline{\vec{0}}} \quad \text{car } \vec{OM} // \vec{T}$$

$$\begin{aligned} \vec{m}_0(\vec{F}) &= \vec{OM} \wedge (-\alpha \vec{v}) \\ &= r \vec{e}_r \wedge (-\alpha r \dot{\theta} \vec{e}_\theta) \\ &= \underline{\underline{-\alpha r^2 \dot{\theta} \vec{e}_z}} \end{aligned}$$

$$\begin{aligned} \vec{m}_0(\vec{P}) &= \vec{OM} \wedge mg(\cos\theta \vec{e}_r - \sin\theta \vec{e}_\theta) \\ &= r \vec{e}_r \wedge (-mg \sin\theta \vec{e}_\theta) \\ &= \underline{\underline{-mgr \sin\theta \vec{e}_z}} \end{aligned}$$

2.

② TMC / point fixe O ds le ref du laboratoire suppose galiléen:

$$\begin{aligned} \frac{d\vec{L}_O}{dt} &= \sum \vec{m}_0(\vec{F}) \\ \vec{0} &= (-\alpha l^2 \dot{\theta} - mgr \sin\theta) \vec{e}_z \end{aligned}$$

$\sin\theta_e = 0$

$\Leftrightarrow \underline{\underline{\theta_e = 0 \text{ ou } \pi}}$ (instable)

$$\begin{aligned}
 2 \quad (3) \quad \vec{L}_0 &= \vec{OM} \wedge m\vec{v} \\
 &= r\vec{u}_r \wedge m\dot{\theta}\vec{u}_\theta \\
 &= mr^2\dot{\theta}\vec{u}_z \\
 &= ml^2\dot{\theta}\vec{u}_z
 \end{aligned}$$

$$\frac{d\vec{L}_0}{dt} = ml^2\ddot{\theta}\vec{u}_z = -\alpha l^2\dot{\theta}\vec{u}_z - mgl\sin\theta\vec{u}_z$$

$$\ddot{\theta} + \frac{\alpha}{m}\dot{\theta} + \frac{g}{l}\sin\theta = 0$$

equation non linéaire
d'ordre 2.

$$2 \quad (4) \quad \theta \ll 1 \text{ rad} \Rightarrow \sin\theta \approx \theta$$

$$\ddot{\theta} + \frac{\alpha}{m}\dot{\theta} + \frac{g}{l}\theta = 0$$

oscill. harmonique
amortie.

$$2 \quad (5) \quad \ddot{\theta} + \frac{\omega_0}{Q}\dot{\theta} + \omega_0^2\theta = 0$$

avec $\omega_0 = \sqrt{\frac{g}{l}}$

$$\frac{\omega_0}{Q} = \frac{\alpha}{m} \Leftrightarrow \frac{1}{Q}\sqrt{\frac{g}{l}} = \frac{\alpha}{m} \Rightarrow \frac{m}{\alpha}\sqrt{\frac{g}{l}} = Q$$

$$2 \quad (6) \quad \text{Régime pseudo-per} \quad Q > \frac{1}{2}$$

$$\frac{m}{\alpha}\sqrt{\frac{g}{l}} > \frac{1}{2}$$

$$2m\sqrt{\frac{g}{l}} > \alpha$$

$$2 \quad (7) \quad \theta(t) = e^{-\frac{\omega_0}{2Q}t} [A \cos \Omega t + B \sin \Omega t] \quad A, B \in \mathbb{R}$$

$$\Omega = \omega_0 \sqrt{1 - \frac{1}{4Q^2}}$$

$$\begin{cases} \theta(0) = 0 \\ \dot{\theta}(0) = l \ddot{\theta}(0) = v_0 \Rightarrow \ddot{\theta}(0) = \frac{v_0}{l} \end{cases}$$

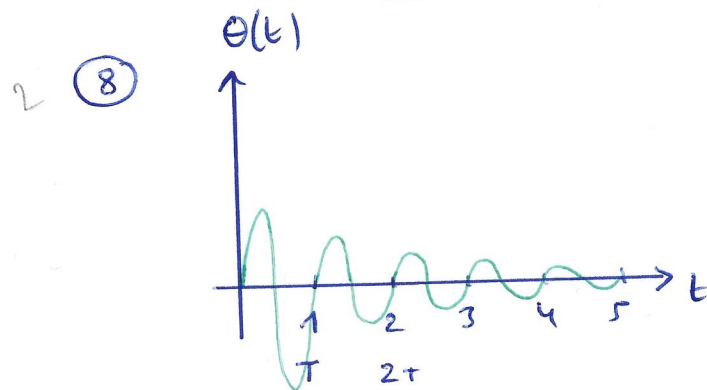
$$\theta(0) = A \cos(0) = A \Rightarrow A = 0.$$

$$\ddot{\theta}(t) = -\frac{\omega_0}{2Q} [B \sin(\Omega t)] + e^{-\frac{\omega_0 t}{2Q}} [\Omega B \cos(\Omega t)]$$

$$\ddot{\theta}(0) = \Omega B \Rightarrow B = \frac{\ddot{\theta}(0)}{\Omega} = \frac{v_0}{l \Omega} = \frac{v_0}{l \omega_0 \sqrt{1 - \frac{1}{4Q^2}}}$$

Final result

$$\theta(t) = e^{-\frac{\omega_0}{2Q} t} \frac{v_0}{l \omega_0 \sqrt{1 - \frac{1}{4Q^2}}} \sin(\Omega t)$$



$$Q = 5.$$

$$T \approx T_0 = \frac{2\pi}{\omega_0}$$

On voit environ 5 oscillations.